



CHHATRAPATI SHAHU JI MAHARAJ UNIVERSITY

Project Synopsis on **XPersonaScan**

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for the Degree of

Bachelors of Computer Application

By:

[PRINCE KATIYAR] [21015001309]

Under the Supervision of
ANKITA GUPTA



PSIT College of Higher Education

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ABSTRACT

In today's rapidly evolving technological landscape, the efficient management of attendance in educational institutions and corporate settings is a pivotal challenge. Traditional methods often fall short, leading to inaccuracies, proxy attendance, and administrative burdens. As a response to these shortcomings, the XPersonaScan Face Recognition Attendance System emerges as a paradigm-shifting solution, harnessing the power of cutting-edge face recognition technology to revolutionize attendance tracking.

The XPersonaScan project is driven by the imperative to address the limitations of conventional attendance systems. By employing advanced computer vision techniques and deep learning algorithms, this system offers a dynamic approach that not only ensures precision and security but also streamlines administrative processes. Through this innovation, the project envisions a future where attendance management becomes a seamless, accurate, and efficient endeavour.

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INTRODUCTION

The XPersonaScan project aims to develop an advanced attendance management system using face recognition technology. Traditional attendance systems are often inefficient and susceptible to inaccuracies. By utilizing face recognition, this system aims to provide a more secure, convenient, and reliable method for recording and managing attendance.

This aims to develop an advanced attendance management system using face recognition technology. To increase the productivity and efficiency of lectures during classes and reduce the burden on faculty, and to overcome the problems of inaccuracy and inefficiency in the traditional system.

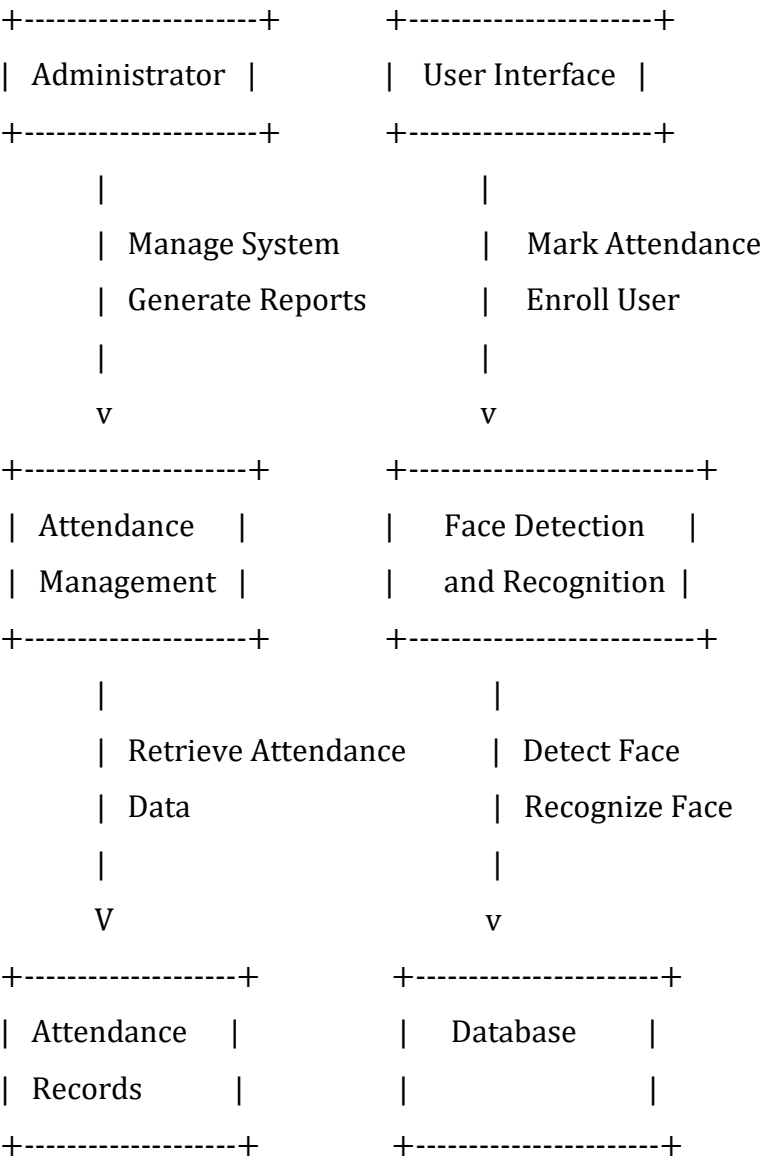
SCOPE

- Provide a flexible system that can be integrated into various educational and corporate environments.
- Continuous improvement of recognition algorithms for better accuracy
- Expansion to handle larger user databases.
- Mobile application development for on-the-go attendance management.

OBJECTIVE

- Implement a face detection and recognition algorithm using Python and relevant libraries
- Develop a user-friendly interface for administrators and users to interact with the system.
- Enhance attendance accuracy and eliminate proxy attendance issues.
- Generate automated attendance reports for easy monitoring and analysis
- Provide a flexible system that can be integrated into various educational and corporate environments.

SYSTEM ARCHTITECTURE



MODULES

TOOLS & TECHNOLOGIES

- Programming Language: Python
- Libraries: OpenCV, TensorFlow, Keras
- User Interface: Tkinter (for desktop application), Flask (for web-based application)
- Database: SQLite, MySQL, or PostgreSQL

SYSTEM REQUIREMENT

Hardware Requirements:-

1. Computer Server:

- A computer or server capable of running the required software components smoothly.
- Sufficient processing power to handle real-time face detection and recognition tasks.
- Ample memory (RAM) to accommodate image processing and deep learning operations.

2. Camera

- A high-resolution camera capable of capturing clear facial images.
- Infrared capabilities for low-light scenarios, if required.
- Positioning to capture faces of users during enrollment and attendance marking accurately

3. Storage

- Adequate storage space for storing the face recognition model, user data, and attendance records.
- Depending on the user base, consider both internal storage and external options.

Software Requirements:-

1. Operating System:

- Windows, Linux, or macOS, depending on the development and deployment environment.
- Compatibility with the chosen programming languages and libraries.

2. Programming Languages:

- Python for developing the system components and algorithms.
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3. Libraries and Frameworks:

- OpenCV: For face detection and image processing.
- TensorFlow or Keras: For developing and training the deep learning model.
- Flask or Tkinter: For building the user interfaces (web-based and desktop).
- NumPy, Pandas: For data manipulation and analysis.
- SQLite, MySQL, or PostgreSQL: For database management and storage.

CONCLUSION

The XPersonaScan project aims to revolutionize traditional attendance systems by implementing a robust face recognition system. This technology-driven solution promises to enhance accuracy, security, and efficiency in attendance management processes for educational institutions and businesses.